Exact answers, no map, no mitigation: user-written circuits through a placement-only OS on IBM Heron r2 devices

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Abstract

A user writes, in plain Qiskit, a circuit whose answer is known exactly. A small operating layer (psf-os) places it on an IBM Heron r2 device by translation, without reading any calibration-quality number, adds its own controls to the same job, and runs it with no error mitigation. We report how close the raw answers come and how that is shown. Two points of the report are of use beyond our own numbers. First, an absolute criterion $|M - M_{\text{exact}}| \leq c$ is biased in both directions: on 5 November a problem that DIFFERS under $c = 0.05$ kept 0.82–0.86 of its signal, as much as a problem that AGREES (0.73–0.85), and a small-answer problem AGREES while keeping only 0.73 of its signal. We therefore print the declared verdict first and the retention $R = M/M_{\text{exact}}$ beside it on every row. Second, every number can be checked: answers are sealed by hash before each run, raw counts and job identifiers are committed, and one script re-scores all 13 device results of this paper from the raw counts and finds every number the same. The results: 22 practice circuits on two devices (all 22 READ on both; 19 and 18 AGREE), a daily record from 5 to 7 November (7 runs and one refusal), and IBM's dual-unitary kicked-Ising chain (arXiv:2411.00765) at $N = 51$ and $N = 71$ sites, where the raw decay rates agree with $\cos 2h$ at all three h (largest $|k - \cos 2h|$: 0.0045 at $N = 51$, 0.0017 at $N = 71$) and every depth up to $t = 24$ (1 715 CZ) is readable. A correction built from the job's own controls, declared in advance, reduced the error on both devices but did not make the answers agree across problems. No quantum advantage is claimed: every answer here is computed exactly on a classical machine.

1 Introduction

1.1 The no-map policy

Most reports of device results choose qubits with the device's daily calibration (the "map"): error rates, coherence times and read-out errors measured by the vendor a few hours earlier. Ordinary computers have no such map, and a user who writes a circuit does not usually consult one. Papers 5 and 6 of this series used placements that read the coupling graph and the list of failed elements only; paper 6 (doi:10.5281/zenodo.23883938) added a self-check layer for rings. This paper carries the same policy to circuits that a user writes on the two-dimensional lattice, and to an entrance, `psf-os run2d`, that takes such a circuit as it is.

The policy, as used throughout:

- placement reads the coupling graph and the failed or masked elements of the day's record, nothing else; no quality number, no search over placements, no selection by simulation;
- no error mitigation of any kind on the reported values (dynamical decoupling and twirling off);
- the device of the day is chosen by its queue (pending jobs), not by its quality;
- the answer of every problem is computed exactly beforehand and sealed by hash; the run is declared before any device record of that day is read.

1.2 What this paper adds

Three kinds of results (sections 4–6): practice problems after published experiments, a daily record of new problems, and IBM's dual-unitary chain at two of its widths. Two parts of the report are of a general kind: the bias of an absolute criterion (section 7) and the way every number can be checked by anyone (section 9). A negative result is reported in full (section 8).

2 The OS's two-dimensional layer

2.1 Placement by translation

A user circuit is written on Heron r2 labels ($20r + c$ for row r and column c, with bridge qubits between rows). Its pattern of touched qubits and couplers is moved by translations in a fixed order (offsets of rows and columns that map the heavy-hex lattice onto itself); the first copy that touches no failed qubit or coupler is used. If no copy qualifies, the entrance refuses and names, per copy, the failed elements in the way. Nothing is ranked.

2.2 Floor, mirror and ring sentinel

Every user circuit goes to the device with two companions in the same job, at the same place: the *floor*, the same circuit without its non-diagonal one-qubit gates (ideal: every qubit reads its input), and the *mirror*, the compiled circuit followed by its inverse (ideal: the input back). One ring sentinel per job, a 40-qubit echo on a fixed ring, reports whether the device as a whole behaved during the job. The mirror is a fault locator, not a measure of the forward error: on 4 November a P4 circuit at $t = 16$ had a mirror mean of 0.05 while its answer agreed within 0.03 (`results/practice-2026-11-04/README.md`).

2.3 The answer card

For each user circuit the entrance writes a card: per measured qubit the value, its two standard errors, the exact value (from the sealed answers, opened after the run), and the floor and mirror means. Every verdict below is computed from these cards by a scorer committed before the run.

2.4 Writing $R_{ZZ}(\pm\pi/2)$ as one CZ

$R_{ZZ}(\pm\pi/2)$ equals one CZ with single-qubit R_Z gates on both sides, exactly. Since psf-os ac84182 the compiler writes it so; before, the default synthesis used two CZ. The practice problems P4 and P5 need half the CZ gates they needed before (`results/practice-2026-11-04/README.md`).

3 Problems and answers

3.1 Problems after published experiments

- **P4**: IBM's dual-unitary kicked-Ising chain (arXiv:2411.00765, SI Eq. S5), $J = \pi/4$, on a 33-site chain (C33) at $b = \pi/4 - 0.3$ and $b = \pi/4 - 0.05$, $t = 2$–16; X read-out.
- **P5**: the form of IBM's utility experiment (Kim et al., Nature 618, 500 (2023)): per step $R_{ZZ}(-\pi/2)$ on every edge, then $R_X(\theta)$ on every qubit; 151 qubits, $\theta \in \{0.4, 0.8, 1.0\}$, $t = 2$–8.
- **P6**: the two-dimensional Ising problem of the Qedma QESEM tutorial (35 qubits, $\theta_x = 0.53$ and two neighbouring values, $t = 1$–9).
- **The daily pool** (`daily/pool.py`): problems D01–D32 in a fixed order and one fixed problem, FX-A (P4's chain, $h = 0.15$, $b = \pi/4 - 0.3$, $t = 12$; the same circuit as the practice point `p4_b0.485_t12`), from three families: F1 (P6's form, on a 32-qubit patch), F2 (P5's rule, on patches of 42–102 qubits) and F3 (P4's chain).

3.2 Sealing

For every run the SHA-256 of the answers file is committed (and pushed) before the device record of that day is fetched; the values are committed after the verdicts. The daily pool's hashes went in at commit 12164be, its values at 50e412e.

3.3 Two independent computations

Each answer of the pool is computed twice by different methods: GPU state vectors or light cones (with four internal checks), and a CPU recomputation by another method (`results/daily-pool-2026-11-05/README.md`). The dual-unitary answers are in closed form, $\langle X_t \rangle = \cos(2h)^t$ on the light-cone site t and 0 elsewhere for $t \le (N-1)/2$, checked against state vectors at $N = 11, 21$ (to 2.5×10^{-13}) and a matrix-product state at $N = 51$ (to 2.6×10^{-12}).

4 Result 1: practice problems (4 November)

22 user circuits went through the entrance on ibm_marrakesh and on ibm_fez (table (a) in the appendix; jobs dblibpd8mdtc73d66v90 and dblin89kbkrc73cnpcbg). Declared verdicts: READS 22/22 on both devices; AGREES 19/22 (marrakesh) and 18/22 (fez); BEATS the better trivial guess 19/22 and 20/22.

The misses have one shape: P5 at $\theta = 0.4$ is short by -0.06 to -0.07 on both devices at every depth from $t = 4$. Its answers are close to 1, so a loss of 6–8 % of the signal is enough to leave the band, and the all-one guess is hard to beat. Its retention there is 0.92–0.94 on both devices, about the same as at $\theta = 0.8$, where the answers agree. On fez the deepest P4 point ($t = 16$) also leaves the band (-0.060, retention 0.61).

5 Result 2: the daily record (5–7 November)

From 5 November one fixed problem (FX-A) and the next problem of the pool run every night on the device with the shortest queue (rules: `prereg/daily/standing_rules.md`). Every night and device is a row, refusals included (tables (b) and (c)).

- 5 November, both devices: FX-A AGREES (-0.024 marrakesh, -0.043 fez); D02 DIFFERS (-0.111, -0.147) while keeping 0.86 and 0.82 of its signal.
- 6 November, marrakesh: D03 (72 qubits) REFUSED; failed qubits 46, 73 and 82 were in every copy. By an amendment declared that morning FX-A ran alone and AGREES (-0.008, retention 0.95).
- 7 November, fez: FX-A AGREES (-0.038; placed at the copy with row offset 4); D04 AGREES ($+0.010$).

FX-A from day to day (table (c)): on fez the retention was 0.74, 0.73 and 0.76 (4, 5 and 7 November), within 0.03; on marrakesh 0.86 and 0.85 (4 and 5 November) and 0.95 on 6 November, when FX-A ran alone in its job. The two devices differ by more than fez varies from day to day.

6 Result 3: IBM's dual-unitary chain at $N = 51$ and $N = 71$

The experiment of arXiv:2411.00765 was run by IBM at the dual-unitary point on ibm_strasbourg at $N = 51, 71, 91$, with 262 144 shots per point and a stack of error mitigation (tensor-network error mitigation, Pauli twirling with noise learning, Clifford-point calibration, informationally complete measurement, read-out mitigation). We ran the same circuit, written as a user would (`gpu/practice_circuits.p4`, $J = b = \pi/4$), through the entrance: open chain, $|+\rangle$ on site 0, Bell pairs on the other sites, t brickwork layers, X read-out; $h \in \{0, 0.05, 0.10, 0.15\}$, $t \in \{4, 8, 12, 16, 20, 24\}$; 4 000 shots per point; no mitigation. The chain is the first qualifying copy of the first qualifying shape in a fixed list; $h = 0$ is the device's own ruler (exact value 1 on the cone). The judgements, declared on 3 November and reused unchanged: READS (mean squared error of the on-cone values at most half the mean square of the exact values); the decay rate k per $h > 0$, fitted to the ratio $\langle X_t \rangle(h)/\langle X_t \rangle(0)$ as k^t, AGREES if $|k - \cos 2h| \le \max(2\,\mathrm{SE}, 0.005)$; and the readable depth (the largest t above 3 SE).

Table 1: Dual-unitary chain on ibm_kingston, raw. k: fitted decay rate (SE). Records: `results/du51-2026-11-07`, `results/du71-2026-11-07` (7 November 2026, CET); the 21-site run of 3 November (job V of `results/du-repro-2026-11-03`) for comparison.

	$N = 21$ (3 Nov.)	$N = 51$	$N = 71$	$\cos 2h$
k, $h = 0.05$	0.9971 (0.0007)	0.9950 (0.0006)	0.9948 (0.0007)	0.9950
k, $h = 0.10$	0.9810 (0.0008)	0.9821 (0.0008)	0.9799 (0.0008)	0.9801
k, $h = 0.15$	0.9573 (0.0011)	0.9598 (0.0011)	0.9570 (0.0012)	0.9553
rate verdicts	AGREES ×3	AGREES ×3	AGREES ×3	
READS: MSE / mean square	0.0123 / 0.845	0.0403 / 0.699	0.0508 / 0.699	
readable depth	—	24 (every h)	24 (every h)	
most CZ in one circuit	—	1 225	1 715	
sentinels	CLEAN	CLEAN, CLEAN	CLEAN, CLEAN	
jobs	dbkjdm1ffo6c73bbl8d0	dbnfljdt7fqc73ajisig, dbnflnk4148s7390vi20	dbnij8c4148s73913t50, dbnijck4148s73913t90	

The raw values themselves decay with depth: at $h = 0$ the on-cone value falls from 0.92 ($t = 4$) to 0.63 ($t = 24$) at $N = 51$, and from 0.94 to 0.50 at $N = 71$. The rate test divides by this ruler, so a loss common to all h cancels; a loss that depended on h would not. The rates agree at every h and both widths; the largest distance to $\cos 2h$ is 0.0045 at $N = 51$ ($h = 0.15$, inside the 0.005 tolerance) and 0.0017 at $N = 71$. The 91-site chain did not fit on that day's records: no shape of the list avoided the failed elements.

What is and is not compared with IBM: we claim the same circuit, raw, from a user's entrance, at two of IBM's widths. We do not claim IBM's depth (their circuits reach 4 095 two-qubit gates at $N = 91$), their accuracy after mitigation, or any ranking. IBM's unmitigated points are shown only as plotted symbols in their figures, so we do not copy them as numbers.

7 Verdict and retention

Every table here prints the declared verdict first and, beside it on every row, the retention $R = M/M_{\text{exact}}$ and the normalised error $|M - M_{\text{exact}}|/|M_{\text{exact}}|$ (not given when $|M_{\text{exact}}| < 0.1$). No row is left out because it looks worse (`notes/reporting/verdict_and_retention.md`).

The reason is a bias that any benchmark scored by $|M - M_{\text{exact}}| \le c$ carries, and it works in both directions:

- *a DIFFERS that keeps as much signal as an AGREES*: on 5 November D02 DIFFERS on both devices yet kept 0.86 and 0.82 of its signal; FX-A, which AGREES, kept 0.85 and 0.73;
- *an AGREES that loses much*: FX-A on fez kept 0.73–0.76 of its signal on every day and AGREES every time, because its answer is small (the same relative loss gives a smaller absolute difference);
- among the practice points, P5 at $\theta = 0.4$ DIFFERS with retention 0.92–0.94, while P5 at $\theta = 0.8$ AGREES with retention 0.89–0.97.

The decay per CZ layer differs by problem family as well (5 November: D02 0.004–0.005, FX-A 0.007–0.013), so we do not claim a single "device health" number.

8 A negative result: correction

The floor and the mirror measure, in the same job, how much signal the device loses without and with the gates. A correction that divides by them was declared on 4 November and tested on P6 on both devices (table (d)). It HELPS everywhere (the error falls to 0.02–0.40 of the raw error), but the corrected answers AGREE in three of six cases only: on fez it over-corrects at $\theta_x = 0.8$ (+0.022 rising to +0.098), and on marrakesh two of three remain outside the band.

An exploratory run on 5 November (not declared, marked so in table (d)) added randomised frames (a CZ twirl and a block twirl). The twirl halved the mirror's over-correction of FX-A, but it cost forward signal: D02 raw went from −0.092 to −0.159 (marrakesh) and from −0.143 to −0.239 (fez). For FX-A every corrected variant had a larger error than the raw value (10 of 10), and 9 of 10 left the band. A correction built on the mirror is problem-dependent here, and none of the variants made the answers agree across problems. All values reported in this paper are raw.

9 Reproduction

Three levels of checking are open to a reader.

1. *Re-score*: `python reproduce.py` re-runs every committed scorer, unchanged, on a copy of each result from its committed raw counts, without any access to IBM, and compares every number with the committed scores. For the 13 device results of this paper (P6 on two devices, the 2D test, the correction on two devices, practice on two devices, the daily record on 5, 6 and 7 November, the dual-unitary chain at $N = 51$ and $N = 71$) it prints SAME for all 13 (TOQB 7733aba).
2. *Recompute the answers*: every answers file is regenerated by committed code, and its hash compared with the hash committed before the run.
3. *Rerun on a device*: the job folders (circuits as QPY, the manifest) are committed with their job identifiers; the circuits can be sent again with `psf-os submit`.

Software versions are listed in `results/ENVIRONMENT.md`.

10 Limits

- Three devices of one family and a few days; the practice and daily runs were made in hours when the floors were good (0.95–0.98).
- The queue rule picks the device with the shortest queue, which is not always a good device: on 4 November it chose fez, whose floor was low that morning.
- An outside comparison that does not favour us: on 4 November our unmitigated P6 differences on fez at $t = 5, 7, 9$ were 0.159, 0.210 and 0.201, against the tutorial's unmitigated 0.091, 0.139 and 0.180 (their gate implementation and shots are not stated; `results/p6-device-2026-11-04/README.md`).
- Some patches fit no copy on a given day (D03 refused on 6 November; the 91-site chain on 7 November).
- Every answer is classically computable; the record shows how a device behaves on problems whose answer is known, not what it can do beyond them.

11 Discussion: distance to the wall (an estimate, not a measurement)

How far are such raw runs from problems of practical interest? We take one example, chemistry, through a number only (no chemistry problem was run). From the kept rows of our records the loss per CZ layer γ has a median of 0.0011 (smallest 0.0005, largest 0.0052; `notes/paper7/wall_distance.md`). Without error correction, a relative error ε allows $L_{\max} = -\ln(1-\varepsilon)/\gamma$ CZ layers: at the median, about 92 layers for $\varepsilon = 0.1$ and fewer than one for $\varepsilon = 0.001$. The published resource estimates for FeMoco (Reiher et al. 2017; Lee et al. 2021) would need, run uncorrected, a γ 5.5 to 13.4 orders of magnitude smaller than ours, depending on the estimate and on how parallel the circuit is taken to be. Error correction changes this conclusion entirely; the estimate only says how far raw runs of the kind reported here are from that regime.

12 Conclusion

A user's circuit, placed without a map and run without mitigation, gives answers that can be compared with exact values, checked by anyone, and read with both the declared verdict and the retained signal. On the practice problems all 22 circuits READ on both devices; in the daily record the fixed problem kept 0.73–0.76 of its signal on fez on every day; and IBM's dual-unitary chain was reproduced raw at 51 and 71 sites with decay rates within 0.005 of $\cos 2h$. A correction built from our own controls did not work across problems. No advantage is claimed.

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Data and code

TOQB (`https://github.com/TN-Holdings-LLC/TOQB`, commit 7733aba) holds the declarations (`prereg/`), the results with raw counts, job identifiers and released answers (`results/`), the scorers (`gpu/`), `reproduce.py` and the tables of this paper (`notes/paper7/tables.py`). psf-os (`https://github.com/TN-Holdings-LLC/psf-os`) is the operating layer; the commit used for each result is listed in `results/ENVIRONMENT.md`.

References

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2. Y. Kim et al., Evidence for the utility of quantum computing before fault tolerance, Nature 618, 500 (2023).
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A Generated tables

Generated by `notes/paper7/tables.py` from the committed records; no number in them is typed by hand.

Table 2: (a) Practice problems, 2026-11-04 (declared; raw, no correction). The declared criterion is an absolute difference; the same relative loss gives a larger difference when the exact answer is larger. R and the normalised error show the loss relative to the answer.

device	problem	READS	AGREES	BEATS	M - M_exact	R	norm. error	job
ibm_marrakesh	p4_b0.485_t02	READS	AGREES	BEATS	-0.005	0.978	0.022	dblibpd8mdtc73d66v90
ibm_marrakesh	p4_b0.485_t04	READS	AGREES	BEATS	-0.006	0.972	0.028	dblibpd8mdtc73d66v90
ibm_marrakesh	p4_b0.485_t08	READS	AGREES	BEATS	-0.021	0.887	0.113	dblibpd8mdtc73d66v90
ibm_marrakesh	p4_b0.485_t12	READS	AGREES	BEATS	-0.022	0.862	0.138	dblibpd8mdtc73d66v90
ibm_marrakesh	p4_b0.485_t16	READS	AGREES	BEATS	-0.031	0.799	0.201	dblibpd8mdtc73d66v90
ibm_marrakesh	p4_b0.735_t02	READS	AGREES	BEATS	+0.024	small answer	small answer	dblibpd8mdtc73d66v90
ibm_marrakesh	p4_b0.735_t04	READS	AGREES	BEATS	+0.015	small answer	small answer	dblibpd8mdtc73d66v90
ibm_marrakesh	p4_b0.735_t08	READS	AGREES	BEATS	+0.013	small answer	small answer	dblibpd8mdtc73d66v90
ibm_marrakesh	p4_b0.735_t12	READS	AGREES	BEATS	+0.005	small answer	small answer	dblibpd8mdtc73d66v90
ibm_marrakesh	p4_b0.735_t16	READS	AGREES	BEATS	+0.001	small answer	small answer	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx0.4_t2	READS	AGREES	DOES NOT BEAT	-0.046	0.950	0.050	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx0.4_t4	READS	DIFFERS	DOES NOT BEAT	-0.063	0.933	0.067	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx0.4_t6	READS	DIFFERS	DOES NOT BEAT	-0.060	0.933	0.067	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx0.4_t8	READS	DIFFERS	BEATS	-0.070	0.922	0.078	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx0.8_t2	READS	AGREES	BEATS	-0.024	0.963	0.037	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx0.8_t4	READS	AGREES	BEATS	-0.030	0.941	0.059	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx0.8_t6	READS	AGREES	BEATS	-0.021	0.951	0.049	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx0.8_t8	READS	AGREES	BEATS	-0.048	0.893	0.107	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx1.0_t2	READS	AGREES	BEATS	-0.010	0.977	0.023	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx1.0_t4	READS	AGREES	BEATS	-0.000	0.998	0.002	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx1.0_t6	READS	AGREES	BEATS	+0.009	1.054	0.054	dblibpd8mdtc73d66v90
ibm_marrakesh	p5_tx1.0_t8	READS	AGREES	BEATS	-0.004	0.976	0.024	dblibpd8mdtc73d66v90
ibm_fez	p4_b0.485_t02	READS	AGREES	BEATS	-0.018	0.917	0.083	dblin89kbkrc73cnpcbg
ibm_fez	p4_b0.485_t04	READS	AGREES	BEATS	-0.027	0.882	0.118	dblin89kbkrc73cnpcbg
ibm_fez	p4_b0.485_t08	READS	AGREES	BEATS	-0.039	0.785	0.215	dblin89kbkrc73cnpcbg
ibm_fez	p4_b0.485_t12	READS	AGREES	BEATS	-0.041	0.740	0.260	dblin89kbkrc73cnpcbg
ibm_fez	p4_b0.485_t16	READS	DIFFERS	BEATS	-0.060	0.612	0.388	dblin89kbkrc73cnpcbg
ibm_fez	p4_b0.735_t02	READS	AGREES	BEATS	+0.014	small answer	small answer	dblin89kbkrc73cnpcbg
ibm_fez	p4_b0.735_t04	READS	AGREES	BEATS	+0.010	small answer	small answer	dblin89kbkrc73cnpcbg
ibm_fez	p4_b0.735_t08	READS	AGREES	BEATS	-0.001	small answer	small answer	dblin89kbkrc73cnpcbg
ibm_fez	p4_b0.735_t12	READS	AGREES	BEATS	-0.013	small answer	small answer	dblin89kbkrc73cnpcbg
ibm_fez	p4_b0.735_t16	READS	AGREES	BEATS	-0.018	small answer	small answer	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx0.4_t2	READS	AGREES	BEATS	-0.041	0.956	0.044	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx0.4_t4	READS	DIFFERS	DOES NOT BEAT	-0.061	0.936	0.064	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx0.4_t6	READS	DIFFERS	DOES NOT BEAT	-0.061	0.933	0.067	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx0.4_t8	READS	DIFFERS	BEATS	-0.056	0.938	0.062	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx0.8_t2	READS	AGREES	BEATS	-0.023	0.965	0.035	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx0.8_t4	READS	AGREES	BEATS	-0.021	0.958	0.042	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx0.8_t6	READS	AGREES	BEATS	-0.012	0.973	0.027	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx0.8_t8	READS	AGREES	BEATS	-0.017	0.961	0.039	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx1.0_t2	READS	AGREES	BEATS	-0.009	0.980	0.020	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx1.0_t4	READS	AGREES	BEATS	+0.006	1.025	0.025	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx1.0_t6	READS	AGREES	BEATS	+0.015	1.096	0.096	dblin89kbkrc73cnpcbg
ibm_fez	p5_tx1.0_t8	READS	AGREES	BEATS	+0.012	1.072	0.072	dblin89kbkrc73cnpcbg

Table 3: (b) Daily record (declared; raw, no correction). The declared criterion is an absolute difference; the same relative loss gives a larger difference when the exact answer is larger. R and the normalised error show the loss relative to the answer.

date	device	problem	AGREES	BEATS	M - M_exact	R	norm. error	decay / CZ layer	job
2026-11-05	ibm_fez	D02	DIFFERS	BEATS	-0.147	0.816	0.184	0.0052	dbm5hid8ft8s738u88g0
2026-11-05	ibm_fez	FX-A	AGREES	BEATS	-0.043	0.732	0.268	0.0130	dbm5hid8ft8s738u88g0
2026-11-05	ibm_marrakesh	D02	DIFFERS	BEATS	-0.111	0.860	0.140	0.0038	dbm5hgd8mdtc73d7bang
2026-11-05	ibm_marrakesh	FX-A	AGREES	BEATS	-0.024	0.846	0.154	0.0070	dbm5hgd8mdtc73d7bang
2026-11-06	ibm_marrakesh	D03	REFUSED	-	-	-	-	-	not run (no copy qualifies)
2026-11-06	ibm_marrakesh	FX-A	AGREES	BEATS	-0.008	0.949	0.051	0.0017	dbmqhbu74qds73dsrde0
2026-11-07	ibm_fez	D04	AGREES	BEATS	+0.010	1.022	0.022	-0.0005	dbn5ludt7fqc73aivq60
2026-11-07	ibm_fez	FX-A	AGREES	BEATS	-0.038	0.763	0.237	0.0121	dbn5ludt7fqc73aivq60

Table 4: (c) FX-A day to day (the same circuit as practice p4_b0.485_t12). The declared criterion is an absolute difference; the same relative loss gives a larger difference when the exact answer is larger. R and the normalised error show the loss relative to the answer.

date	device	M - M_exact	R	AGREES	job
2026-11-04 (practice)	ibm_marrakesh	-0.022	0.862	AGREES	dblibpd8mdtc73d66v90
2026-11-04 (practice)	ibm_fez	-0.041	0.740	AGREES	dblin89kbkrc73cnpcbg
2026-11-05	ibm_fez	-0.043	0.732	AGREES	dbm5hid8ft8s738u88g0
2026-11-05	ibm_marrakesh	-0.024	0.846	AGREES	dbm5hgd8mdtc73d7bang
2026-11-06	ibm_marrakesh	-0.008	0.949	AGREES	dbmqhbu74qds73dsrde0
2026-11-07	ibm_fez	-0.038	0.763	AGREES	dbn5ludt7fqc73aivq60

Table 5: (d) Corrections compared (declared 11-04; exploratory 11-05 marked so: corrected MSE over the qubits kept by rule R, raw over all; AGREES after computed here with the same 0.05). The declared criterion is an absolute difference; the same relative loss gives a larger difference when the exact answer is larger. R and the normalised error show the loss relative to the answer.

kind	device	problem	raw MSE	corrected MSE	ratio	verdict	AGREES after	job
declared (11-04)	ibm_fez	P6 theta_x 0.53	0.0419	0.0034	0.082	HELPS	AGREES	dblgrsm74qds73dqfbqg
declared (11-04)	ibm_fez	P6 theta_x 0.3	0.0439	0.0008	0.018	HELPS	AGREES	dblgrsm74qds73dqfbqg
declared (11-04)	ibm_fez	P6 theta_x 0.8	0.0205	0.0083	0.403	HELPS	DIFFERS	dblgrsm74qds73dqfbqg
declared (11-04)	ibm_marrakesh	P6 theta_x 0.53	0.0897	0.0169	0.189	HELPS	DIFFERS	dblgv8pkbkrc73cnn6qg
declared (11-04)	ibm_marrakesh	P6 theta_x 0.3	0.0875	0.0092	0.105	HELPS	DIFFERS	dblgv8pkbkrc73cnn6qg
declared (11-04)	ibm_marrakesh	P6 theta_x 0.8	0.0450	0.0072	0.161	HELPS	AGREES	dblgv8pkbkrc73cnn6qg
exploratory, not declared (11-05, run)	ibm_fez	D02 corr_raw	0.0343	0.0117	0.341	not judged	AGREES	dbm6471kbkrc73cou7e0
exploratory, not declared (11-05, run)	ibm_fez	D02 corr_tw	0.0343	0.0316	0.921	not judged	AGREES	dbm6471kbkrc73cou7e0
exploratory, not declared (11-05, run)	ibm_fez	FX-A corr_raw	0.0027	0.0085	3.104	not judged	DIFFERS	dbm6471kbkrc73cou7e0
exploratory, not declared (11-05, run)	ibm_fez	FX-A corr_tw	0.0027	0.0073	2.665	not judged	AGREES	dbm6471kbkrc73cou7e0
exploratory, not declared (11-05, run2)	ibm_fez	D02 corr_raw	0.0372	0.0133	0.357	not judged	AGREES	dbm6jfd8mdtc73d7dit0
exploratory, not declared (11-05, run2)	ibm_fez	D02 corr_tw	0.0372	0.0346	0.929	not judged	AGREES	dbm6jfd8mdtc73d7dit0
exploratory, not declared (11-05, run2)	ibm_fez	D02 corr_btw	0.0372	0.0296	0.796	not judged	AGREES	dbm6jfd8mdtc73d7dit0
exploratory, not declared (11-05, run2)	ibm_fez	FX-A corr_raw	0.0048	0.0325	6.755	not judged	DIFFERS	dbm6jfd8mdtc73d7dit0
exploratory, not declared (11-05, run2)	ibm_fez	FX-A corr_tw	0.0048	0.0222	4.610	not judged	DIFFERS	dbm6jfd8mdtc73d7dit0
exploratory, not declared (11-05, run2)	ibm_fez	FX-A corr_btw	0.0048	0.0198	4.109	not judged	DIFFERS	dbm6jfd8mdtc73d7dit0
exploratory, not declared (11-05, run)	ibm_marrakesh	D02 corr_raw	0.0125	0.0043	0.343	not judged	AGREES	dbm644hkbkrc73cou76g
exploratory, not declared (11-05, run)	ibm_marrakesh	D02 corr_tw	0.0125	0.0044	0.357	not judged	AGREES	dbm644hkbkrc73cou76g
exploratory, not declared (11-05, run)	ibm_marrakesh	FX-A corr_raw	0.0030	0.0168	5.609	not judged	DIFFERS	dbm644hkbkrc73cou76g
exploratory, not declared (11-05, run)	ibm_marrakesh	FX-A corr_tw	0.0030	0.0065	2.164	not judged	DIFFERS	dbm644hkbkrc73cou76g
exploratory, not declared (11-05, run2)	ibm_marrakesh	D02 corr_raw	0.0155	0.0032	0.207	not judged	AGREES	dbm6jcl8mdtc73d7dio0
exploratory, not declared (11-05, run2)	ibm_marrakesh	D02 corr_tw	0.0155	0.0053	0.346	not judged	AGREES	dbm6jcl8mdtc73d7dio0
exploratory, not declared (11-05, run2)	ibm_marrakesh	D02 corr_btw	0.0155	0.0055	0.359	not judged	AGREES	dbm6jcl8mdtc73d7dio0
exploratory, not declared (11-05, run2)	ibm_marrakesh	FX-A corr_raw	0.0021	0.0268	12.499	not judged	DIFFERS	dbm6jcl8mdtc73d7dio0
exploratory, not declared (11-05, run2)	ibm_marrakesh	FX-A corr_tw	0.0021	0.0091	4.249	not judged	DIFFERS	dbm6jcl8mdtc73d7dio0
exploratory, not declared (11-05, run2)	ibm_marrakesh	FX-A corr_btw	0.0021	0.0105	4.870	not judged	DIFFERS	dbm6jcl8mdtc73d7dio0